

Systems of inequalities



1 A, C, B

2

a Dashed line

b Solid line

c Solid line

d Dashed line

e Dashed line

f Solid line

g Solid line

h Dashed line

3

a No

b No

c Yes

d No

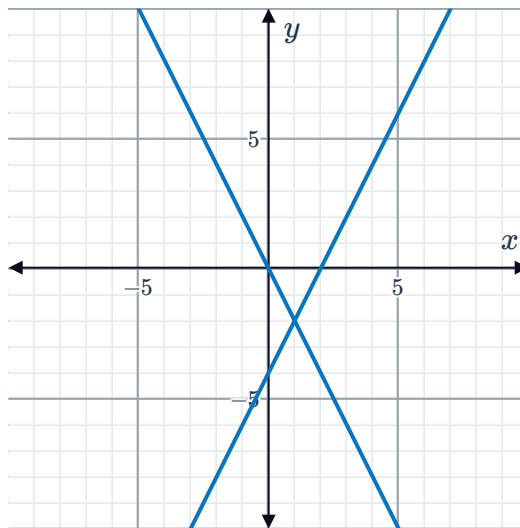
4 The set of a ordered pairs that satisfy each inequality in the system.

5

a

	$2x - y = 4$	$6x + 3y = 0$
x -intercept	(2,0)	(0,0)
y -intercept	(0,-4)	(0,0)

b



c

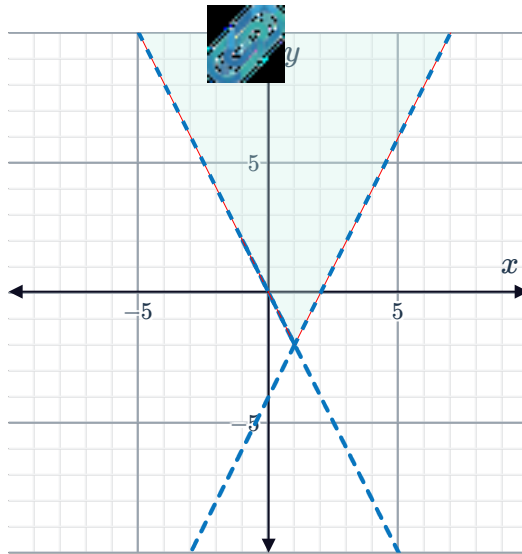
i No

ii Yes

iii No

iv No

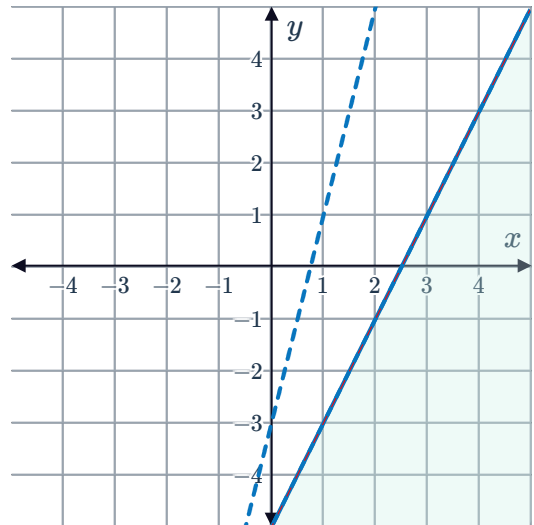
d



6

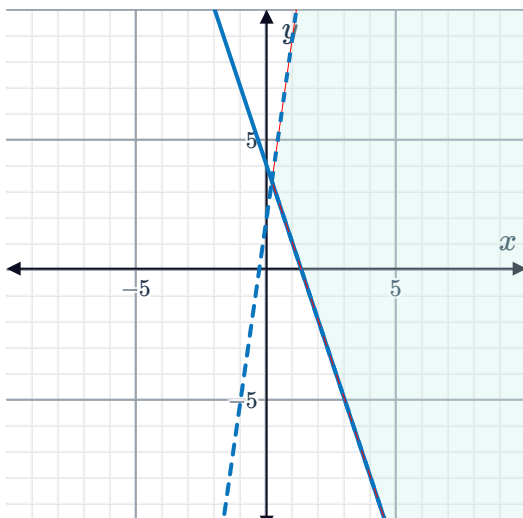
a $(-1, -7)$

b



7

a



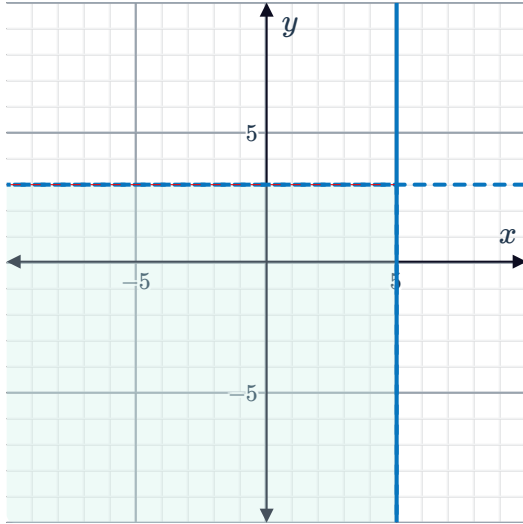
b

- i No
- ii No
- iii No
- iv No
- v Yes
- vi Yes

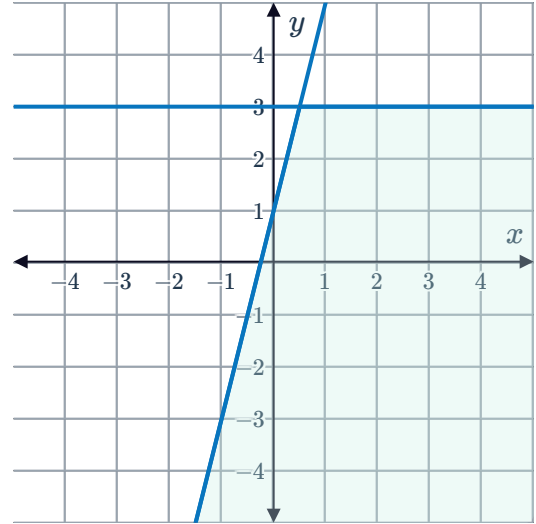


8

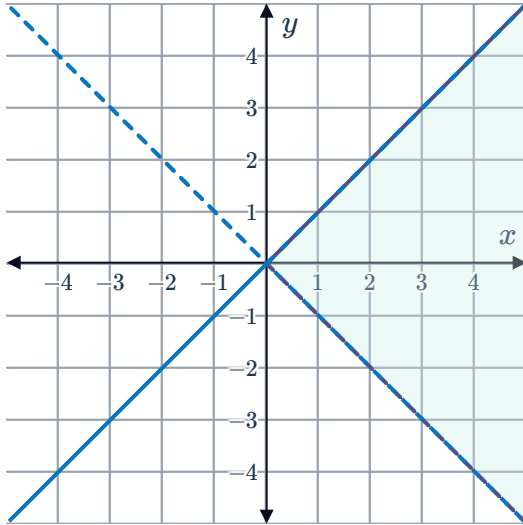
a



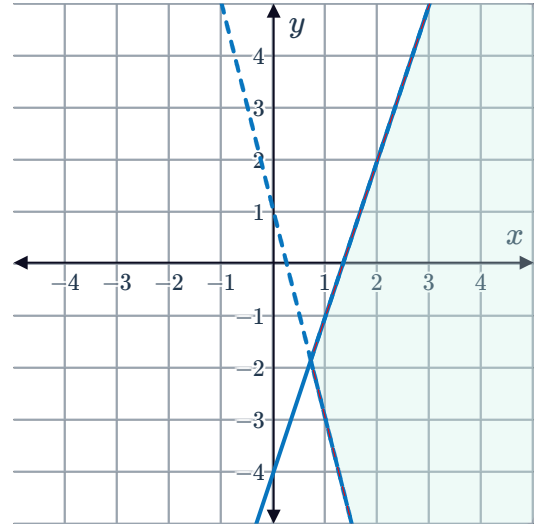
b



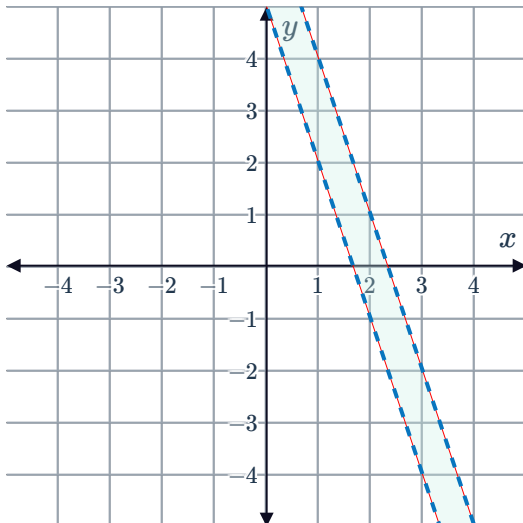
c



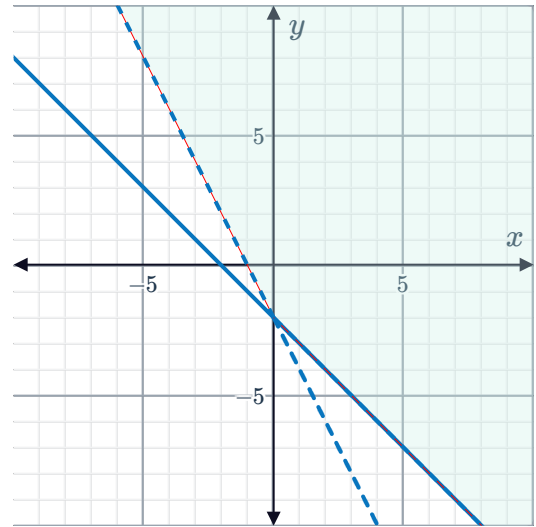
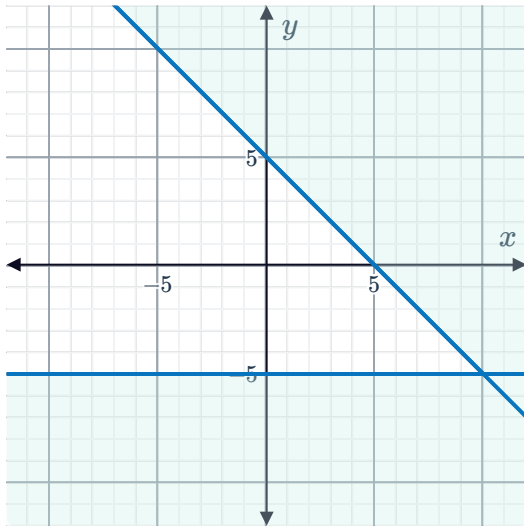
d



e



9 a



10

a No solutions

b Infinitely many solutions

11

a
$$\begin{cases} x < 1 \\ y \leq 3 \end{cases}$$

b
$$\begin{cases} y \geq 2x - 3 \\ y \leq 1 \end{cases}$$

c
$$\begin{cases} y > x \\ y \leq -x \end{cases}$$

d
$$\begin{cases} y + 3x > 0 \\ y + 3x \leq 3 \end{cases}$$

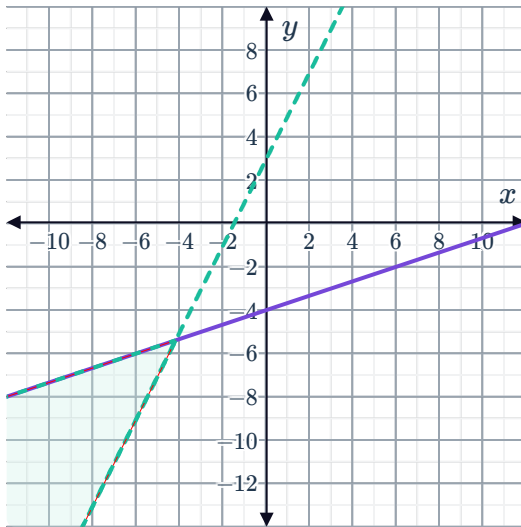
e
$$\begin{cases} x < 1 \\ y < 1 \\ y \geq -x - 5 \end{cases}$$



Additional questions

12

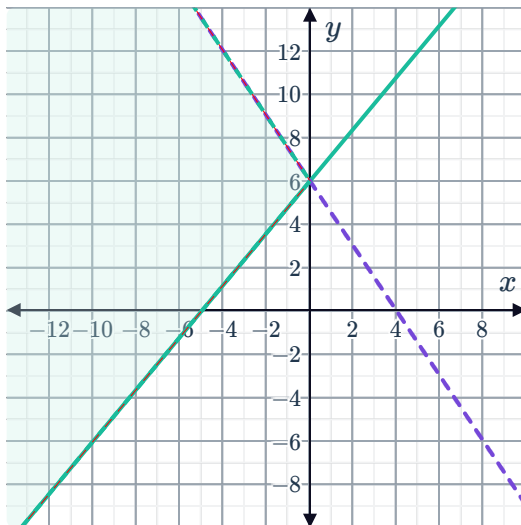
a



b Yes

13

a



b No

14

a

$$x \geq 3, y < \frac{1}{3}x - 1$$

No, since both systems include at least one strict inequality, their point of intersection is not a valid solution.

15 $x + y \leq 15, 0.8x \leq 32$



16 $b + m \leq 27, 10b + 13m \geq 260$

17

a $2v \leq m, 0.6v \leq 42$

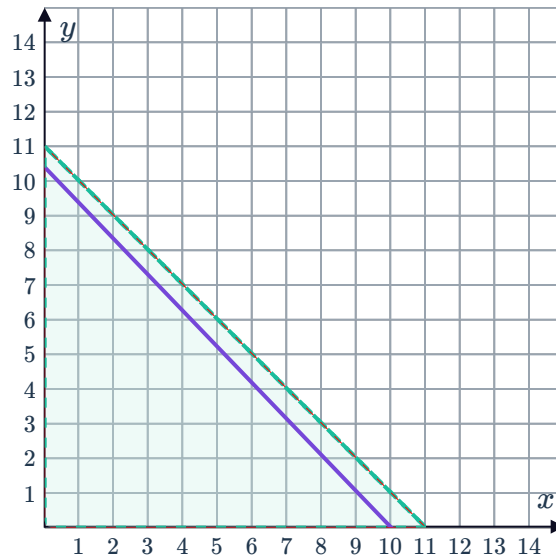
b Yes, since 28 vegan pizzas and 70 meat lovers pizzas would satisfy the constraints, and pizzas can only be made in whole numbers.

c No. While $(25.25, 71)$ satisfies the constraints, you cannot cook a fraction of a pizza.

18

a $w + m \leq 11, 2.6w + 2.5m < 26$

b



c Viable solution: Any point which satisfies the constraints and is made up of positive integers, since the variables represent the number of men and women. For example, $(3, 4)$.

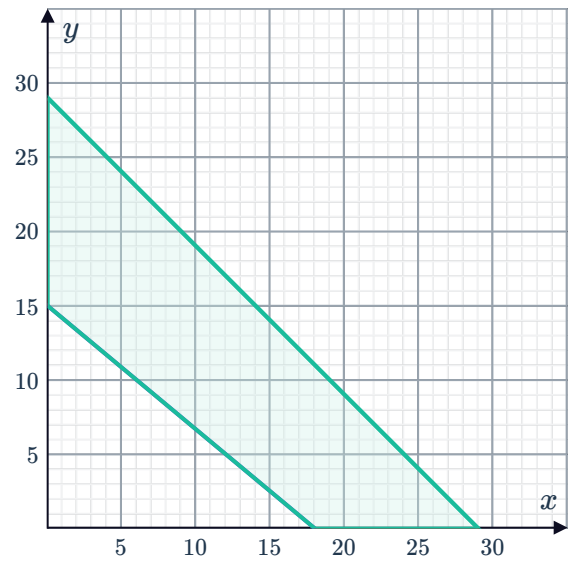
Nonviable solution: Any point which includes at least one non integer value, since you cannot have part of a person as a team member. For example, $(3.1, 4)$.

19

a $x + y \leq 29, 15x + 18y \geq 270$



b



c 10

d Yes